

CONTENTS

TECHNOLOGY

- 25** TRENDS: What To Expect From The Advanced Oscilloscopes
- 32** INTERVIEW: “We Have Achieved A Milestone By Having Government School Children Participate In Championships Alongside Their Private School Counterparts” — Payal Manan Rajpal, Board Member and South Asia Head, Robotex International
- 35** INNOVATION: The Magical, Super-Thin, Flexible Batteries
- 46** INTERVIEW: “MRAM Can Keep Data Even When Power Is Off!” — Sanjeev Aggarwal, President & CEO, Everspin
- 50** INNOVATION: Reflective LCD Technology: Driving Sustainability In Digital Displays



MOVES

- 53** INTERVIEW: “We Ensured Customers’ Monthly EMI Would Be Equal To Or Less Than Their Electricity Savings” — N.P. Ramesh, COO & Co-Founder, Orb Energy
- 55** INTERVIEW: Can MSMEs Contribute And Profit From The Semicon Boom? — Sanjay Gupta, CDO & Global Head of Engineering, L&T Semiconductor Technologies



STRATEGY

- 68** MY STORY: “Miles To Go Before I Sleep” — Prof. (Dr) Rajat Moona, Director, IIT Gandhinagar
- 76** INTERVIEW: Charge Forward: High-Voltage Batteries And MSMEs Can Fuel India’s EV Revolution — Dr Yashodhan P. Gokhale, Vice President for Battery Technology, JSW Energy



DESIGN

- 79** FUTURE: Wi-Fi 6 And Wi-Fi 7 Powering The Next Wave Of Smart Connectivity
- 83** EMBEDDED: Redefining High-Performance Computing In Compact Spaces
- 86** CIRCUIT: Interesting Reference Designs Of LED Control Solution



DO-IT-YOURSELF

- 88** PROJECT: IoT-Based Distribution Transformer Condition Monitoring System
- 92** PROJECT: Flame-Sensing Fire Alarm Using An Arduino Nano
- 95** PROJECT: World’s Smallest Programable IndusPhone Design (Part 2)
- 99** PROJECT: ESP32 Speech Function: Text To Speech
- 101** PROJECT: A Sewer Drain Shield For Smart Cities
- 103** CIRCUIT IDEA: STK435 IC-Based Stereo Amplifier



REGULARS

- 8** Research & Innovation Updates
- 16** Truly Innovative Electronics
- 18** New Products
- 44** Electroons 2.0
- 58** Dashboard: Key Industry Trends That Can Shape The Future Of Your Business
- 60** Startups & Innovators
- 64** Industry News: Industry Updates

EDITOR : Rahul Chopra

EDITORIAL CORRESPONDENCE : Phone: +91-9811155335
E-mail: editor@efy.in

SUBSCRIPTIONS & MISSING ISSUES : Phone: +91-9811155335
E-mail: support@efy.in

BACK ISSUES, BOOKS, KITS, COMPONENTS, etc : Phone: +91-9811155335
E-mail: support@efy.in

NEWSSTAND DISTRIBUTION : Phone: +91-9811155335
E-mail: efycirc@efy.in

ADVERTISEMENTS
NEW DELHI (HEAD OFFICE) : Phone: +91-9811155335
E-mail: efyenq@efy.in

MUMBAI : E-mail: rmowest@efy.in

BENGALURU : E-mail: rmosouth@efy.in

CHINA : Worldwide Focus Media
E-mail: china@efy.in

GERMANY : PMS Plantenberg Media Service GmbH
E-mail: germany@efy.in

JAPAN : Tandem Inc.
E-mail: japan@efy.in

TAIWAN : J.K. Media
E-mail: taiwan@efy.in

UNITED KINGDOM : ASA Media
E-mail: uk@efy.in

UNITED STATES : E & Tech Media
E-mail: usa@efy.in

Printed, published and owned by Ramesh Chopra. Printed at M.P. Printers, B-220, Phase-2, NOIDA-201305, Uttar Pradesh, on the first day of each month and published from D-87/1, Okhla Industrial Area, Phase-1, New Delhi 110020. Copyright 2023. All rights reserved throughout the world. Reproduction of any material from this magazine in any manner without the written permission of the publisher is prohibited. Although every effort is made to ensure accuracy, no responsibility whatsoever is taken for any loss due to publishing errors. EFY will not be responsible for any wrong claims made by an advertiser. Disputes, if any, will be settled in a New Delhi court only.

SUBSCRIPTION RATES

EFY = Electronics For You

PRINT MAGAZINE	1 YEAR (12 copies each)	3 YEARS (36 copies each)	5 YEARS (60 copies each)
WITHIN INDIA (IN RUPEES)			
EFY (Rs 100/copy)	840	2150	3000
SAARC COUNTRIES (IN US\$ BY AIR)			
EFY	50	135	NA
OTHER COUNTRIES (IN US\$ BY AIR)			
EFY	100	270	NA

Courier delivery within India @ Rs 50 per copy

e-magazine subscriptions within India are available at half the rates mentioned here.

Overseas rates for each e-magazine in US\$:

1 year: \$12; 3 years: \$33; 5 years: \$50 only

Please send payments only in favour of **EFY Enterprises Pvt Ltd** through demand draft/cheque/money order to D-87/1, Okhla Industrial Area, Phase 1, New Delhi 110020, or online at: <http://subscribe.efyindia.com> or scan the QR code.

